

Hospital Database Creation and Data Migration

Problem Statement: Hospital Database Migration & Automation

Background: Our hospital has been maintaining all its records—including patient details, doctor rosters, appointments, prescriptions, lab reports, and billing—using Excel files. As our operations have scaled, this method has become inefficient, error-prone, and difficult to manage. We are now transitioning to a relational database system to improve data integrity, performance, and scalability.

Problem Description

- Build a robust, well-structured relational database system that captures all core functionalities of the hospital. From the current Excel-based system, determine the required tables and migrate the data while ensuring data integrity and consistency.
- The database should also support business rules governing appointment management, doctor access to patient data, and generation of department-wise monthly revenue reports.

Problems You Need to Solve with Your Database Design

1. Lack of Unique Identifiers

- Currently, there are no guaranteed unique IDs for patients, doctors, departments, or appointments.
- Introduce something that ensures uniqueness.

2. Disconnected Relationships

- In Excel, appointments are listed, but there is no enforceable link to valid patients or doctors.
- Implement measures to maintain referential integrity between patients, doctors, departments, and appointments.

3. Invalid or Ambiguous Data Entries

- Examples include gender values like 'X', appointment statuses such as 'On Hold', and inconsistent date formats.
- Allowed values: Gender = M/F/O; Status = Scheduled/Completed/Cancelled.

4. Unregulated Scheduling

- Doctors are occasionally double-booked, and appointments are scheduled in the past.
- Prevent invalid appointment entries automatically during insertion.

5. Open Access to Sensitive Patient Information

- All doctors currently see all data regardless of their role or department.
- Implement role-based access so only senior doctors can view all patients in their department; other doctors can access only their respective patients' records.

6. Disconnected Reporting

- There is no way to generate billing or departmental summaries across the hospital.
- Generate monthly revenue reports by department.